# Midterm 2 study guide

**How to prepare — Lectures 18–25 and Quizzes 5–9**

Last updated: November 5, 2026.

Exam: Friday, November 13 · in class day page. Practice problems: Practice packet. Schedule: Schedule.

# Recommended study plan

1. **Scan the learning outcomes below.** Mark each as: confident / shaky / blank.
2. **Go through the lecture notes (18–25).** For each outcome you marked shaky or blank, find the matching fill-in, worked example, or Check yourself item. (Lecture 26 on Nov 9 is review; Lecture 27 is the exam.)
3. **Build your own "important definitions and formulas" list.** As you read, write down every definition or formula you might need on the exam (matrix sizes, $Ax$, RREF language, $A^k$, inverse tests, rank / null / col, determinants).
4. **Rework examples from lecture and quizzes.** Cover the solution, redo Warm-up / Worked / Check yourself items by hand, then uncover and check using the filled in notes..
5. **Start memorizing your list of definitions and formulas.**
6. **Try the practice midterm problems with no notes.**
7. **Check answers** (end of the practice packet). For every miss, update your definitions/formulas list and redo a similar lecture example.
8. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

The exam is **definitions, concepts, and hand computation**, and has no software syntax.

## Useful Course materials

| Source | Role |
|---|---|
| Lectures 18–25 | Core ideas and skills (Unit II through determinants) |
| Practice packet | Definitions, easy computations, applied problems |
| Boyd & Vandenberghe, *VMLS* | Textbook (free online) |
| Strang §3.5 | Rank, column space, null space (selected portions) |

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## Learning outcomes (Lectures 18–25)

### Light review — Quiz 5 / basis (Lecture 14)

1. Explain what a basis accomplishes and what dimension means.
2. Identify a basis and dimension in simple examples.
3. Use span, independence, basis, and dimension accurately when they reappear with matrices (column space, null space, rank).

## Lecture 18 — Matrices as data and structured objects

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

## Lecture 19 — Matrix-vector multiplication

1. Compute $Ax$ by hand for a small matrix.
2. Interpret $Ax$ as a linear combination of the columns of $A$.
3. Interpret $A$ as an input-output transformation.
4. Explain how the dimensions of $A$ determine the input and output spaces.

## Lecture 20 — Transforming space

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply a transform matrix to given points by hand.
3. Interpret a matrix as a coordinate or physical-space transformation.
4. Sketch or describe the image of simple shapes under a stated transform.

## Lecture 21 — Systems, $Ax = b$, and elimination

1. Translate equations into $Ax = b$ and classify the system as square, overdetermined, or underdetermined.
2. Interpret $Ax = b$ as asking whether $b$ is in the span of the columns; anticipate no solution, one solution, or infinitely many.
3. Match each elementary row operation with its matrix, use those matrices to eliminate, and solve a small system by back substitution.
4. Explain why those row operations preserve the solution set.

## Lecture 22 — RREF and linear dynamical systems

1. Read a reduced row echelon form: name pivot and free variables and write a parametric solution.
2. Tell inconsistent, unique, and infinitely many solutions apart from that form.
3. Write an evolving system as $x_{t+1} = Ax_t$ and build the transition matrix from a flow description.
4. Compute and interpret the next state or two.

## Lecture 23 — From row operations to matrix products

1. Apply an elementary matrix to an augmented matrix and read the solution set (including a parametric form).
2. Classify unique / none / infinitely many from RREF and write infinite families with a parameter $t$.
3. Build a transition matrix from a two-compartment flow and compute one step $x_1 = Ax_0$.
4. Explain why two steps need $A(Ax_0)$ / $A^2$, and compute a small product $AB$ by hand.
5. Give an example with $AB \neq BA$, and compute $A^2$ and a small power $A^n x$.

## Lecture 24 — Inverses, rank, and solvability

1. Explain why reducing $[A \mid I]$ to $[I \mid B]$ gives $B = A^{-1}$, and use an inverse to solve $Ax = b$.
2. Interpret $C(A)$ as possible outputs and $N(A)$ as inputs sent to zero.
3. Find rank from pivots and use $\dim N(A) = n - \operatorname{rank}(A)$.
4. Explain when $Ax = b$ is solvable and when a solution is unique.
5. Distinguish square invertibility from full column rank for a tall matrix.

## Lecture 25 — Determinants, geometry, and invertibility

1. Compute determinants of $2 \times 2$ and simple triangular matrices.
2. Interpret determinant magnitude as area or volume scaling.
3. Connect determinant zero with lost dimension and noninvertibility.
4. Recognize the determinant of a triangular matrix from its diagonal.

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